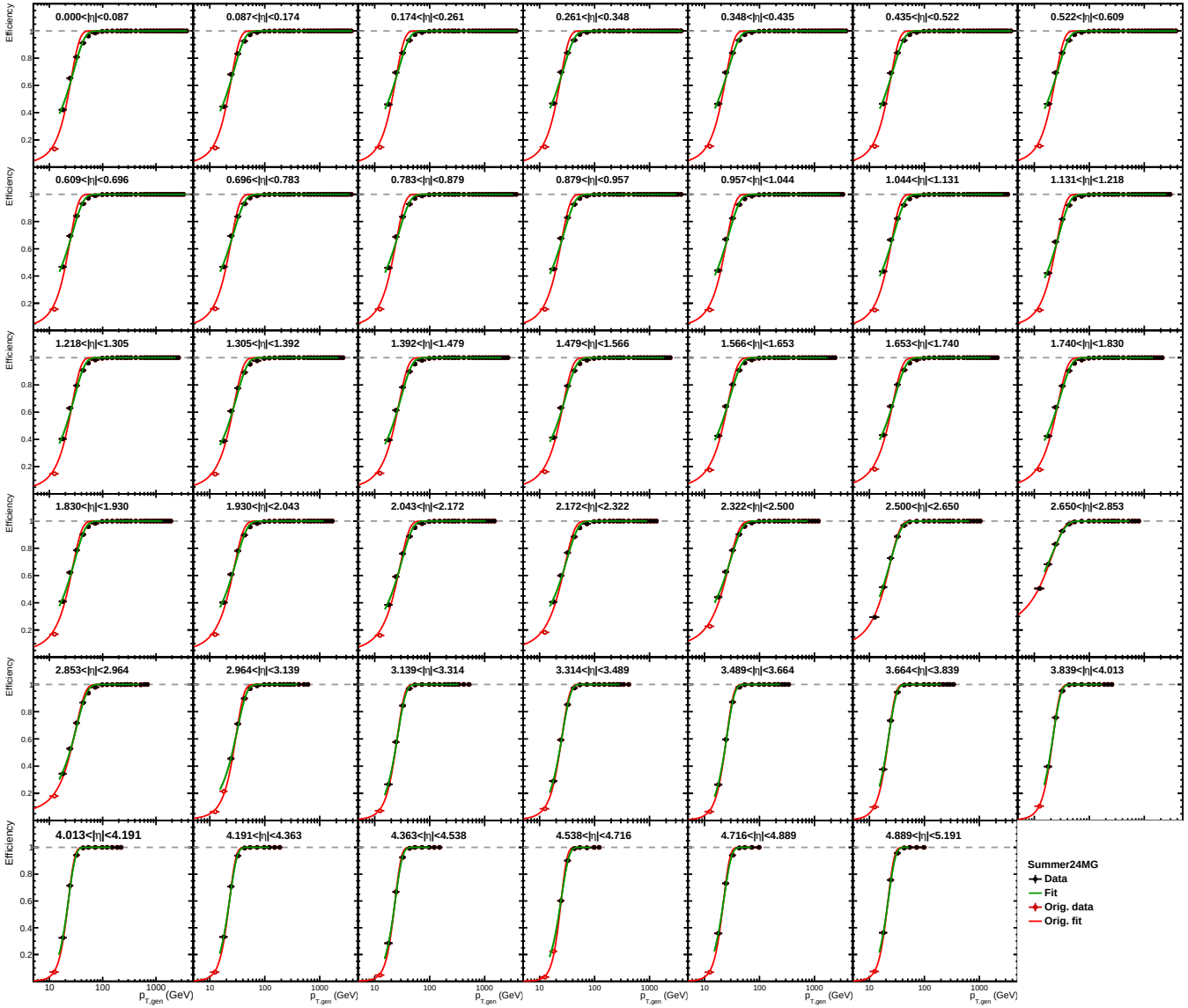
MCTruth L2Relative+JER (MC: Summer24MG, data: Winter25MG)

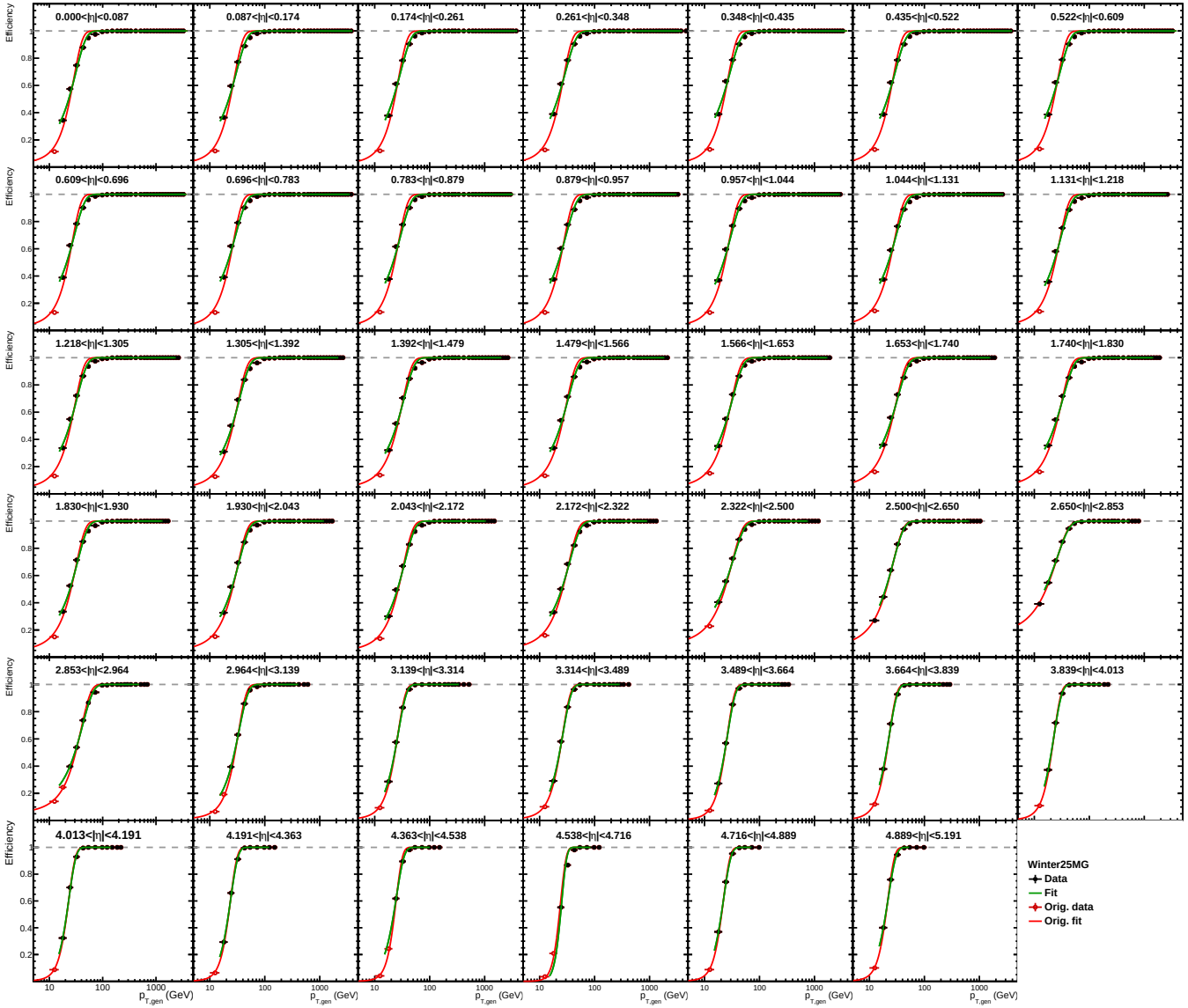
Date: January 16, 2026

- Data residual corrections are derived on top of Winter25MG MC JECs relative to Summer24MG MC with Summer24MG MC JECs
- This hybrid procedure is ok: MC needs to be corrected with it's own MC JEC so that remaining method biases are separated from JEC effects. Data can be corrected with any MC JEC and it makes sense to use the closest matching MC JEC to minimize residual corrections on top of that MC JEC
- Reason for this hybrid method is that Winter25MG are POG samples while Summer24MG has an extensive set of PAG samples, which currently makes Summer24MG more useful for physics analysis

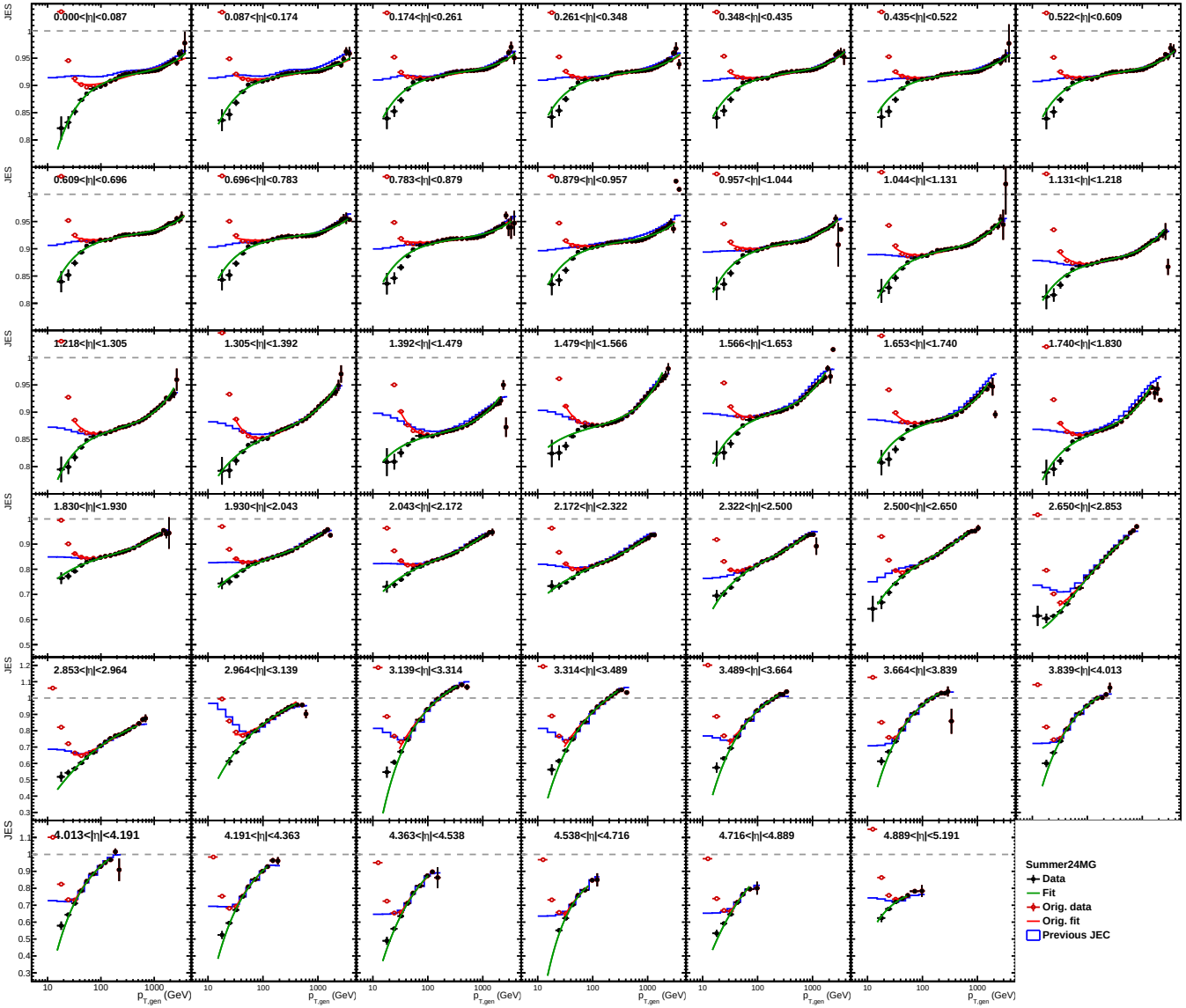
MC efficiency (Summer24MG)



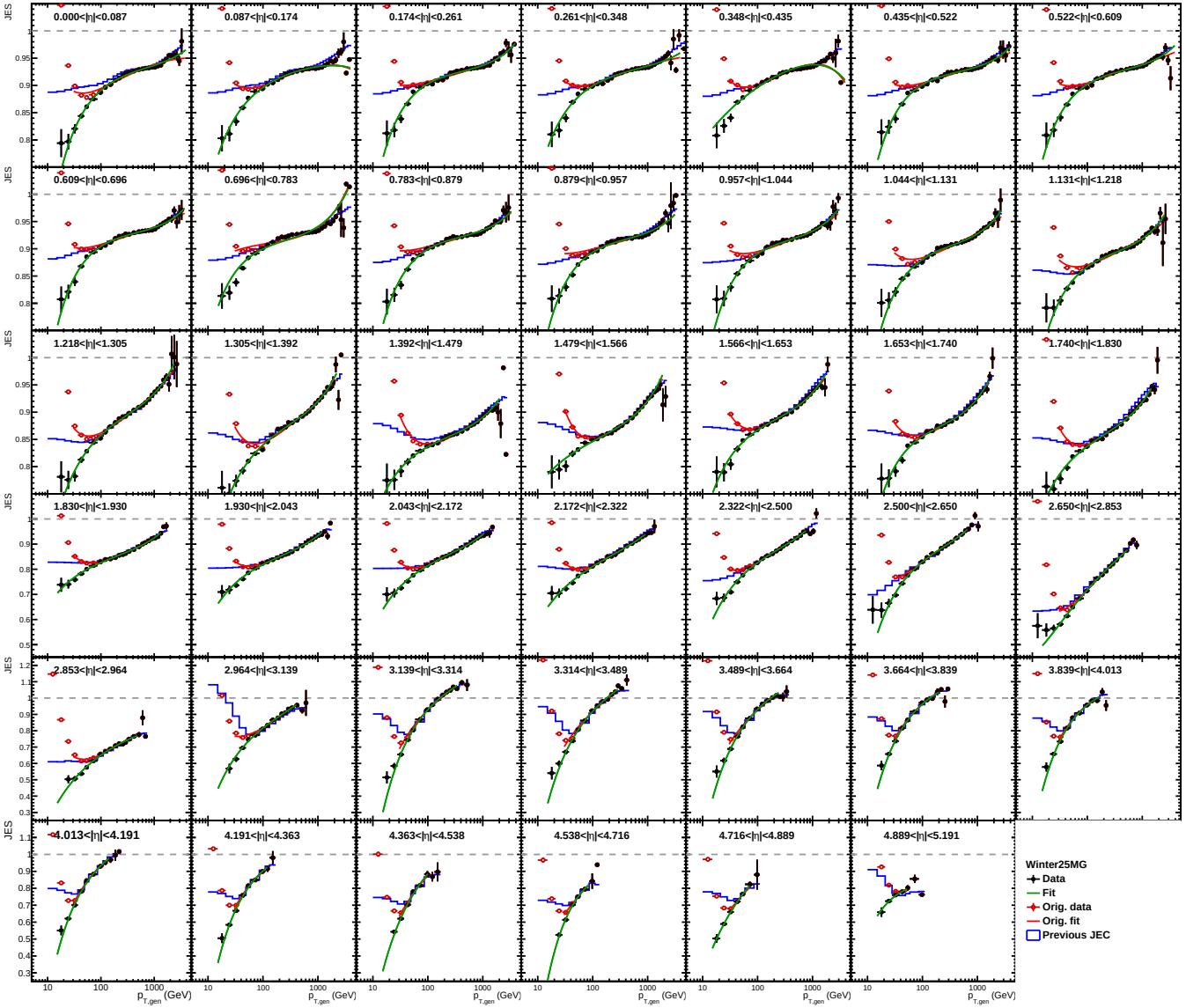
MC efficiency (Winter25MG)



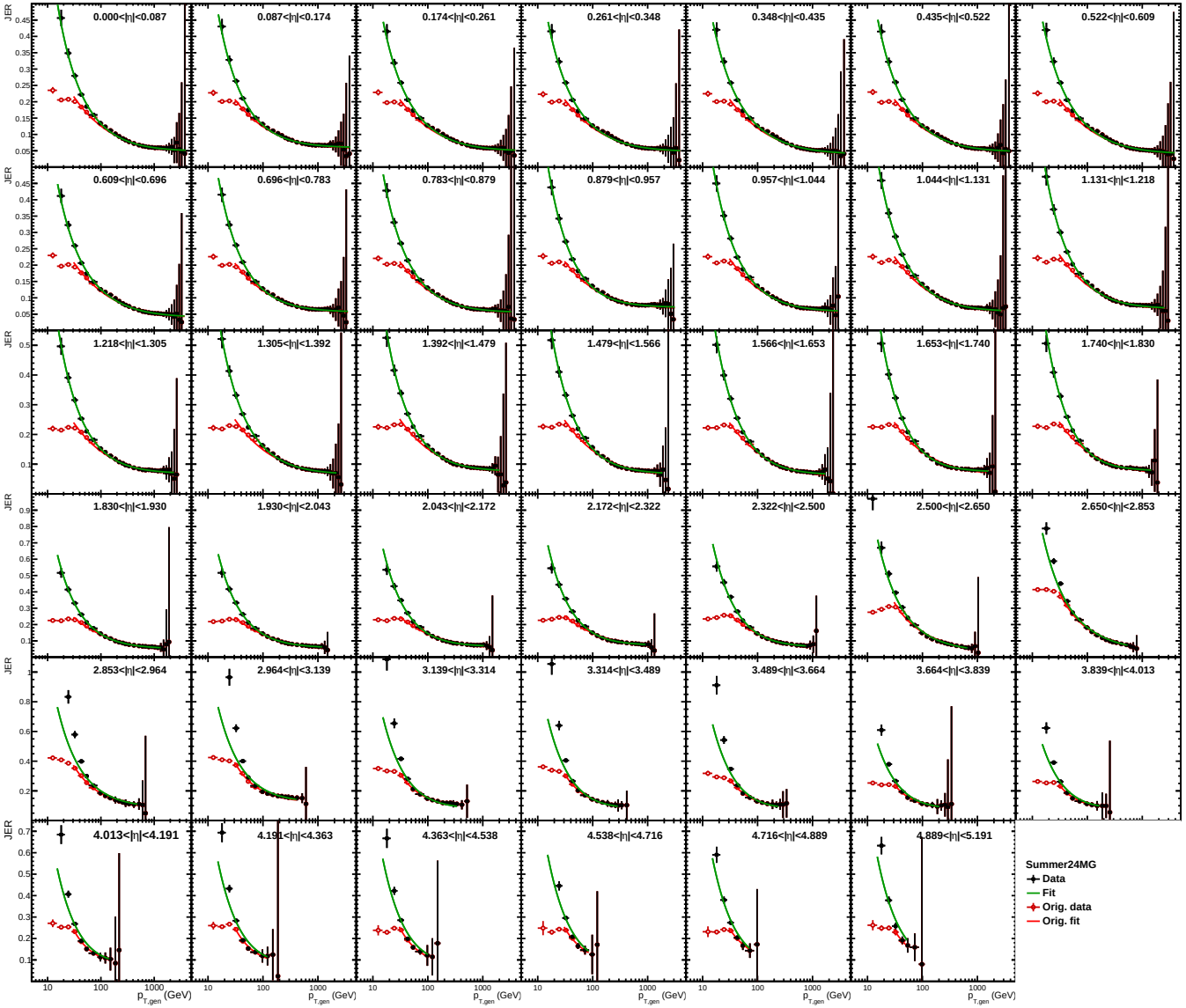
MC JEC (Summer24MG)



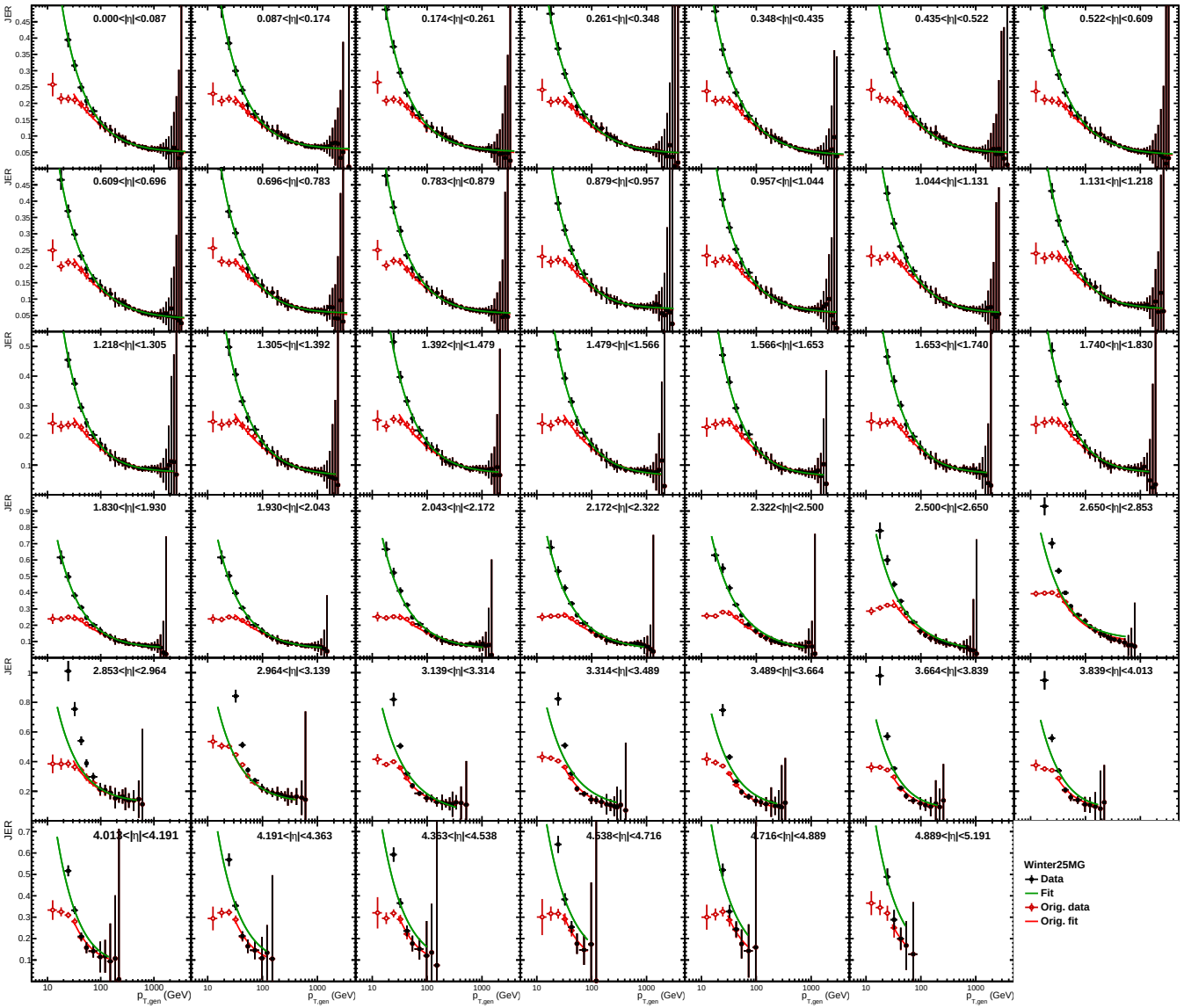
MC JEC (Winter25MG)



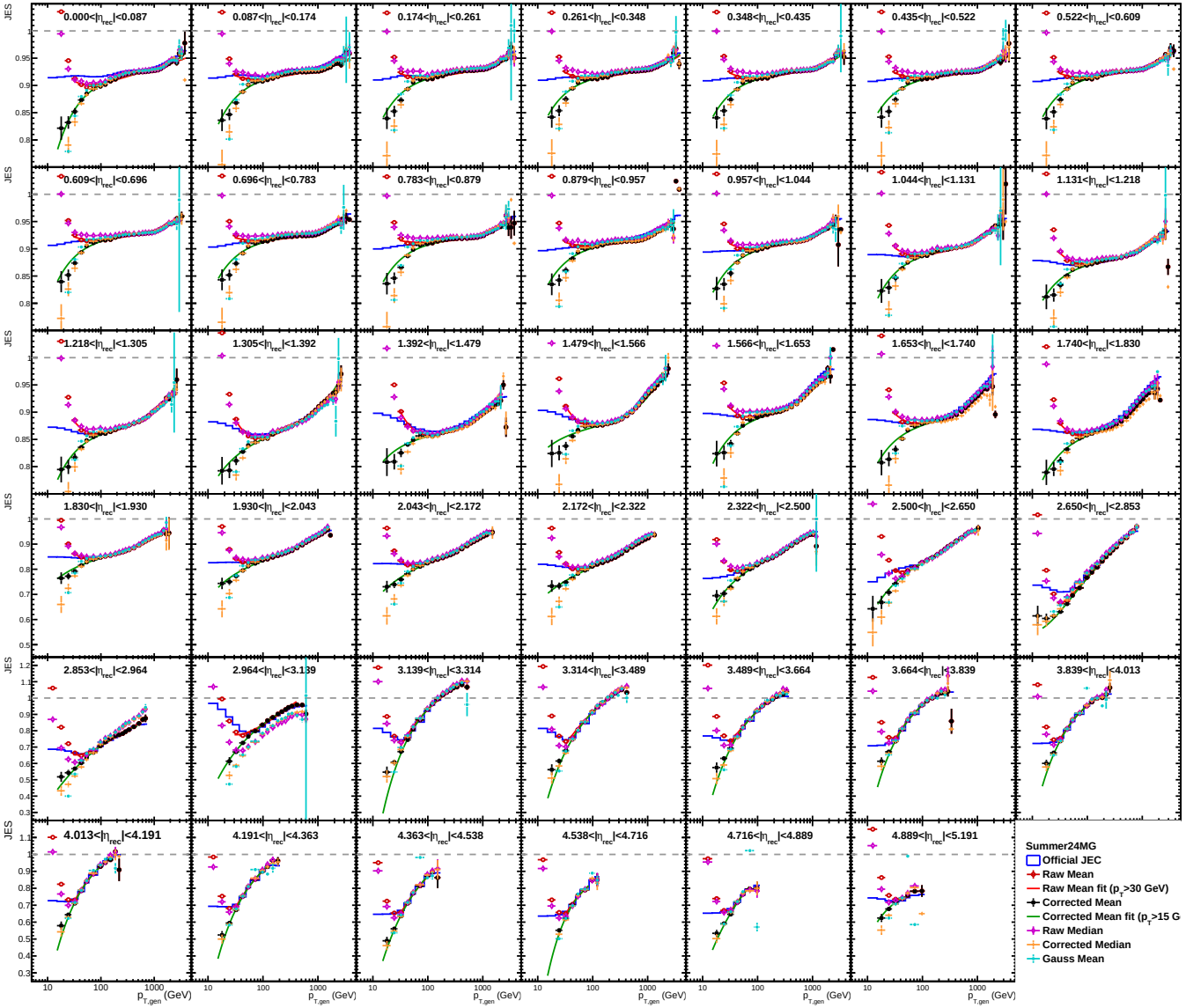
MC JER (Summer24MG)



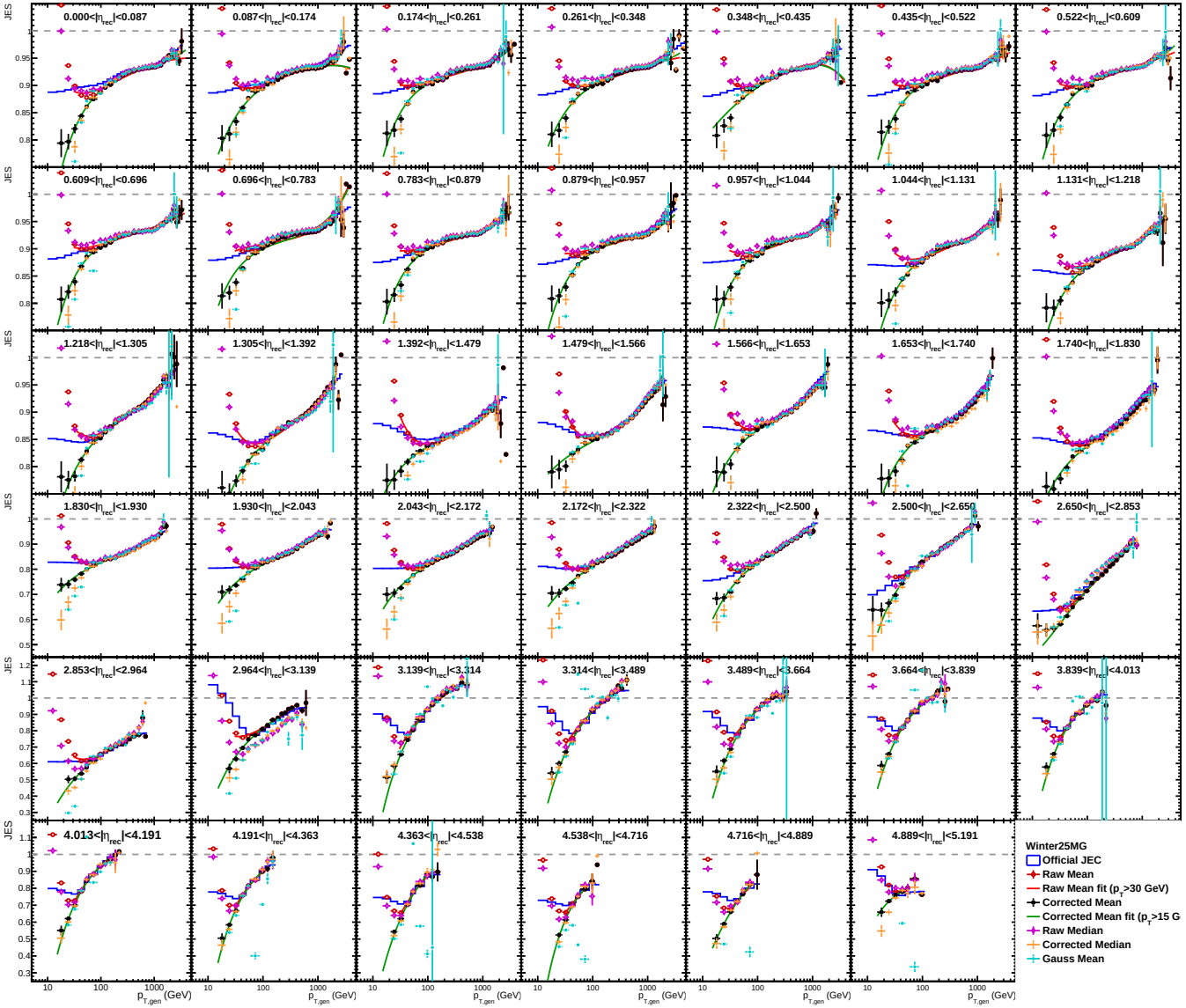
MC JER (Winter25MG)



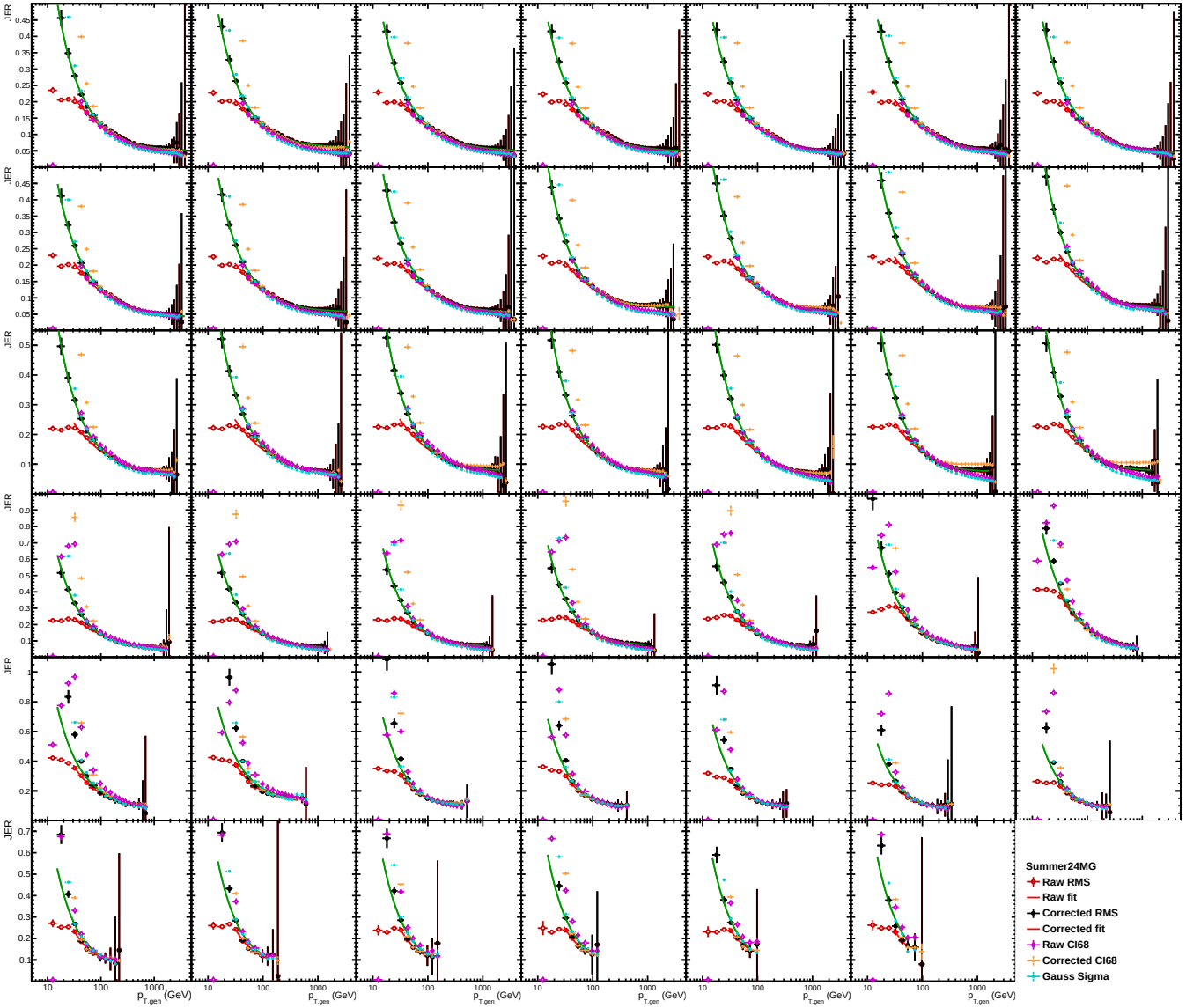
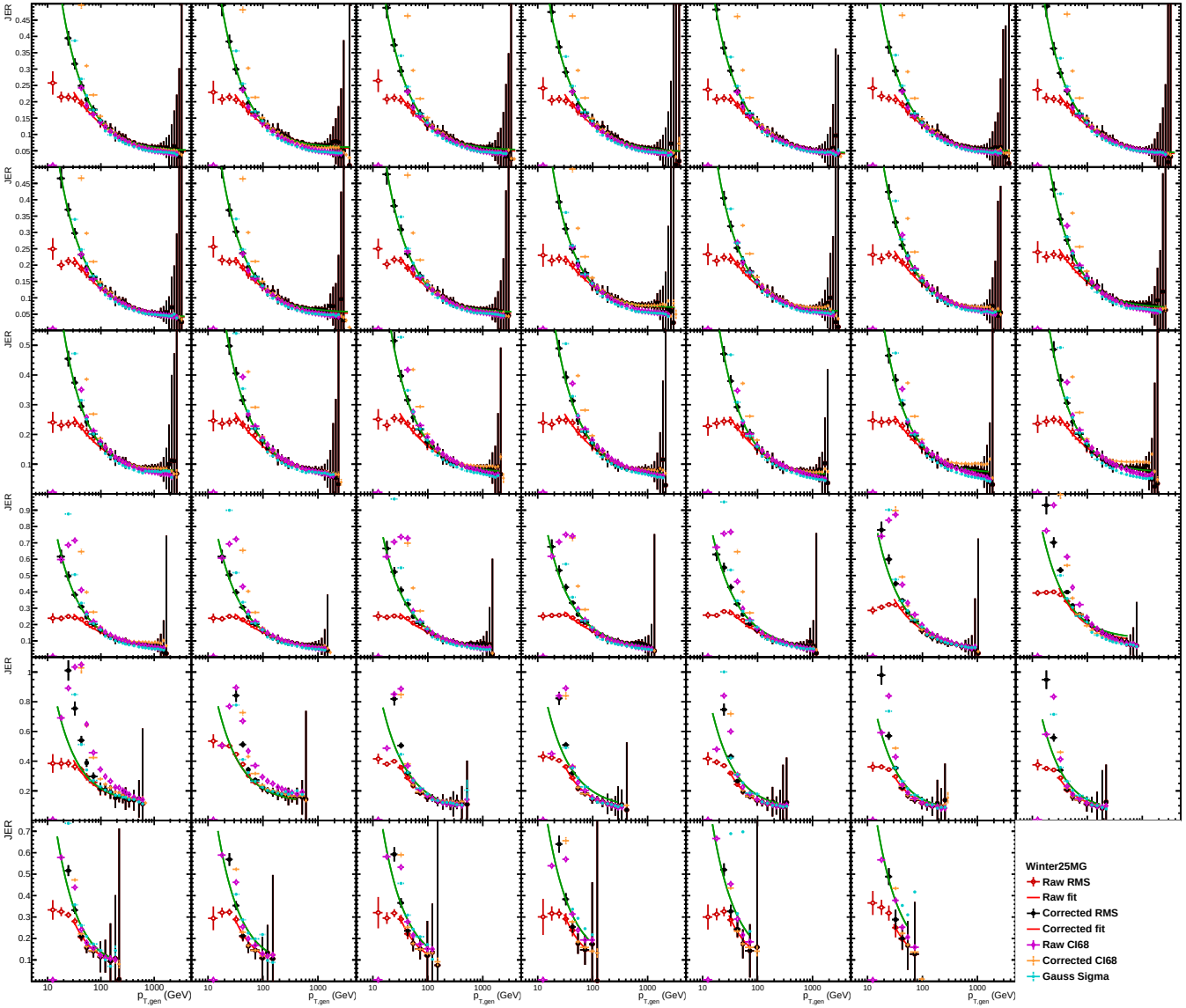
MC JEC extras (Summer24MG)



MC JEC extras (Winter25MG)



MC JER extras (Summer24MG)



MC JER extras (Winter25MG)